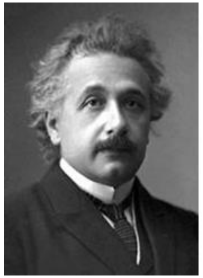
Einstein Podolsky Rosen

Can quantum-mechanical description of **physical reality** be considered **complete**?

A. Einstein, B. Podolsky, and N. Rosen. Can quantum-mechanical description of physical reality be considered complete? Phys. Rev., 47:777, 1935.

Realism

If, without in any way **disturbing** a system,
we can predict with certainty (i.e., with probability equal to unity)
the value of a physical quantity,
then there exists an **element of physical reality**
corresponding to this physical quantity.

A. Einstein, B. Podolsky, and N. Rosen. Can quantum-mechanical description of physical reality be considered complete? Phys. Rev., 47:777, 1935.

Mermin Peres Sqaure

$A_i \in \pm 1$

$R_1 = A_1 \cdot A_2 \cdot A_3 = +1$

$R_2 = A_4 A_5 A_6 = +1$

$R_3 = A_7 A_8 A_9 = +1$

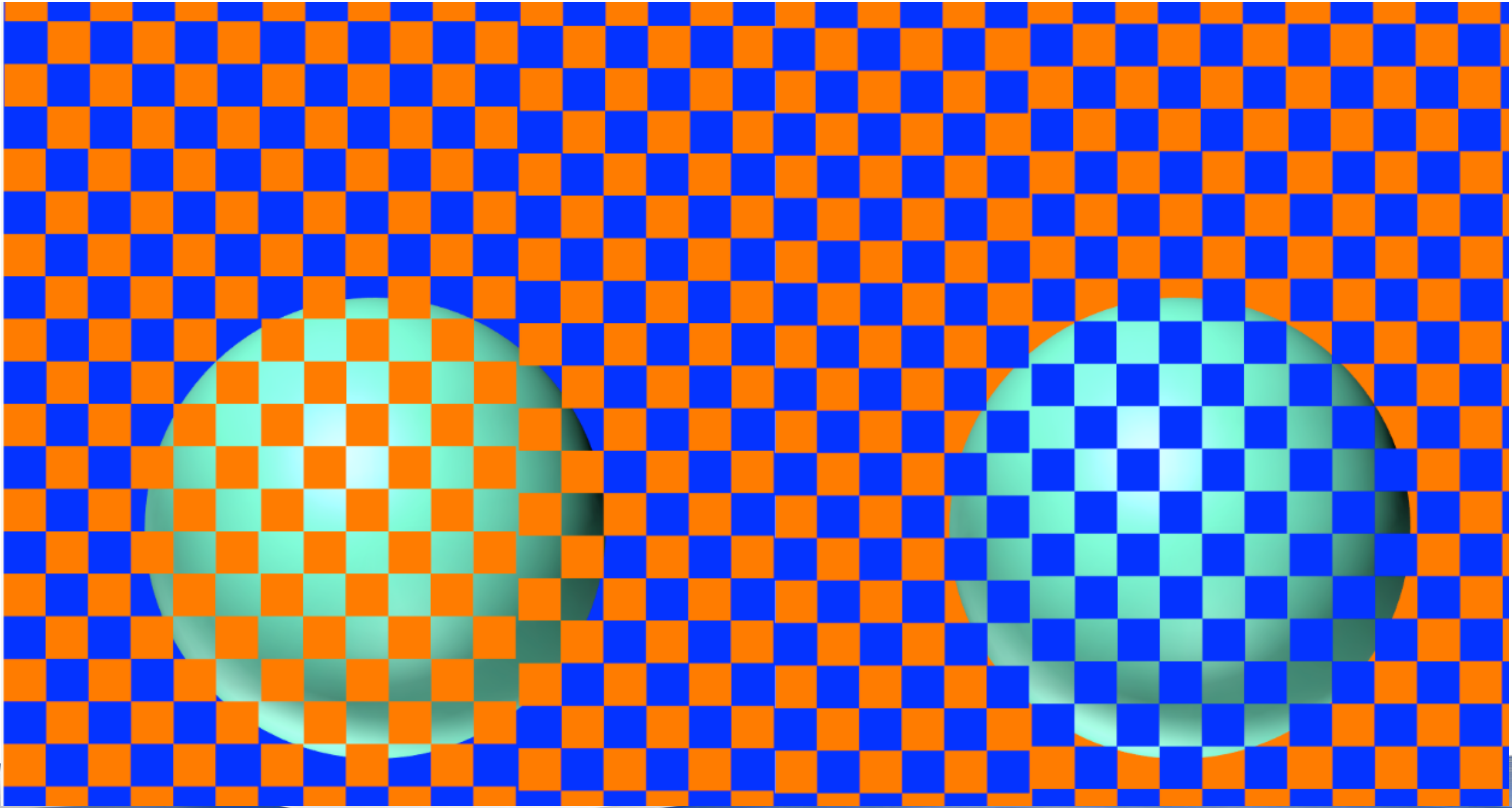
$\vdots$

$C_3 = A_3 A_6 \cdot A_9 = -1$

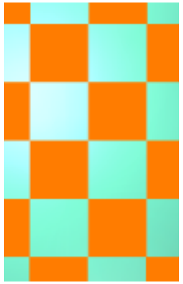
$$M = \prod_{i=1}^{9} A_i = R_1 R_2 R_3 = +1$$

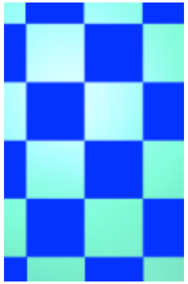
$$\prod_{i=1}^{9} A_i = C_1 \cdot C_2 \cdot C_3 = -1$$

	C1	C2	C3	
R1	A_1	A_2	A_3	1
R2	A_4	A_5	A_6	1
R3	A_7	A_8	A_9	1
	1	1	-1	?



Munker Illusion





Project 1

Using Munker or a similar illusion is it possible to develop a visualization of the Mermin Peres square ?

↳ i.e. a 2-coloring of the 9 cells, that satisfy constraints

	C1	C2	C3	
R1	A_1	A_2	A_3	1
R2	A_4	A_5	A_6	1
R3	A_7	A_8	A_9	1
	1	1	-1	?